SLEEP PARAMETERS ANALYSIS VIA EMPATICA "EMBRACEPLUS" WEARABLE DEVICE

THE WORK IS BEEN DIVIDED IN FOUR MAIN STEPS:

- DATA ACQUISITION
- RAW DATA MANAGEMENT AND STORAGE
- FEATURES EXTRACTION (FROM DIFFERENT TYPES OF SENSORS)
- STATISTICAL ANALYSIS (PRINCIPAL COMPONENT ANALYSIS)

DATA ACQUISITION

DATA FROM 9 DIFFERENT NIGHTS HAVE BEEN ACQUIRED TO SEE IF AND HOW THE APPROACH TO THE EXAM CAN IMPACT SLEEP PARAMETERS.

THE ACQUISITION ROUTINE CAN BE SPLITTED IN THREE PARTS:

- PRE-SLEEP TAG MARK
- SLEEPING PART
- POST-SLEEP TAG MARK

DATA MANAGEMENT

- CONVERTION OF THE ACQUIRED DATA FROM ".AVRO" TO ".MAT" THROUGH A COLAB PYTHON SCRIPT
- DATA SAVING IN RAW_DATA STRUCT CONTAINING VARIOUS INFOS
- SUBDIVISION OF ALL THE DATAS IN NIGHTS, OBTAINED BY THE CONCATENATION OF TWO SUBSEQUENT RAW DATA

Fields	ACC	PPG	EDA	tags	fs_ACC	fs_PPG	fs_EDA	start	t_ACC_datetime	t_PPG_datetime	t_EDA_datetime	date	countup
1	1846560x3 ...	1846464x1 ...	115408x1 d...	1x1 datetime	64	64	3	1732727824	1846560x1 datetime	1846464x1 datetime	115408x1 datetime	'2024-11-27'	1
2	4925408x3 ...	4925120x1 ...	307840x1 d...	2x1 datetime	64	64	4	1732756676	4925408x1 datetime	4925120x1 datetime	307840x1 datetime	'2024-11-28'	2
3	5544768x3 ...	5544768x1 ...	346548x1 d...	1x1 datetime	64	64	3	1732843435	5544768x1 datetime	5544768x1 datetime	346548x1 datetime	'2024-11-29'	3

FEATURES EXTRACTION

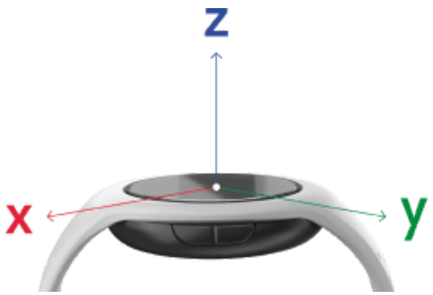
THREE DIFFERENT TYPES OF DATA HAVE BEEN ACQUIRED:

- ACCELERATION DATA: USED TO DETECT SLEEP/AWAKE STATUS
- PPG DATA: FROM WHICH INFOS ABOUT HEART RATE VARIABILITY CAN BE DERIVED
- EDA DATA: DETECTION OF EVENT SUCH AS EDA STORMS

ACCELERATION PROCESSING

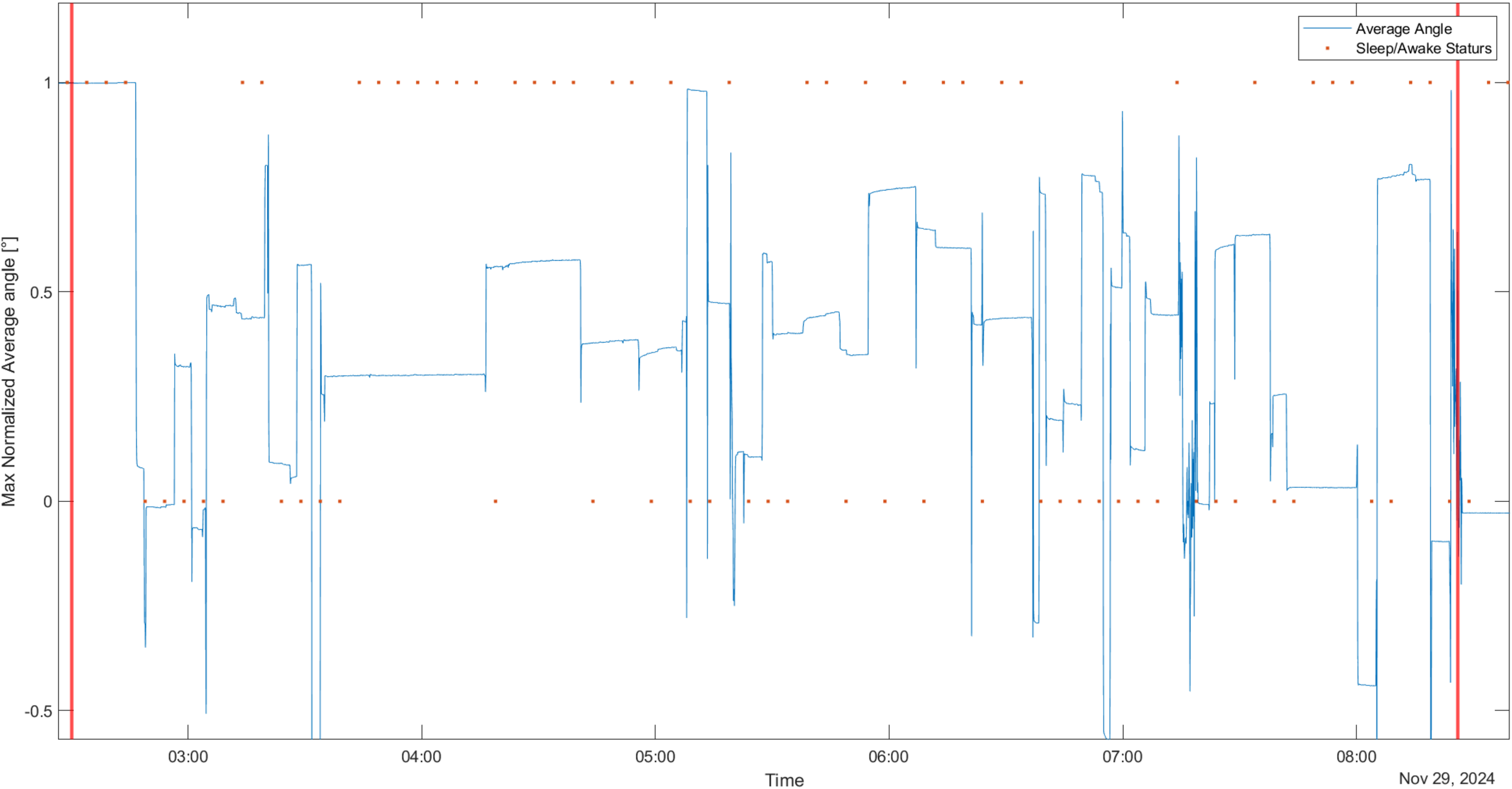
IN ORDER TO DETECT SLEEP/AWAKE STATUS THE VAN HEELS'S ALGORITHM IS APPLIED.

THE VERTICAL ACCELLERATION IS ASSUMED TO BE THE ONE WITH THE INITIAL VALUE NEAREST TO 1 G ($9.81\ m/s^2$)



A VALUE EQUAL TO ZERO OR ONE (WAKE OR ASLEEP) IS THEN ASSIGNED AND WITH ITS CORRESPONDANT REFERENCE DATETIME IS THEN STORED INTO A CELL STRUCTURE

28/29


Average Angle
Sleep/Awake Staturs
Max Normalized Average angle [°]
1
0.5
0
-0.5
03:00
04:00
05:00
06:00
07:00
08:00
Time
Nov 29, 2024

SLEEP EFFICIENCY

AFTER THE SLEEP/AWAKE STATUS DETECTION WE CAN THEN COMPUTE THE SLEEP TIME TO OBTAIN THE SLEEP EFFICIENCY AS:

$$Sleep\ efficiency = \frac{(time\ asleep)}{(total\ time\ in\ bed)} \times 100$$

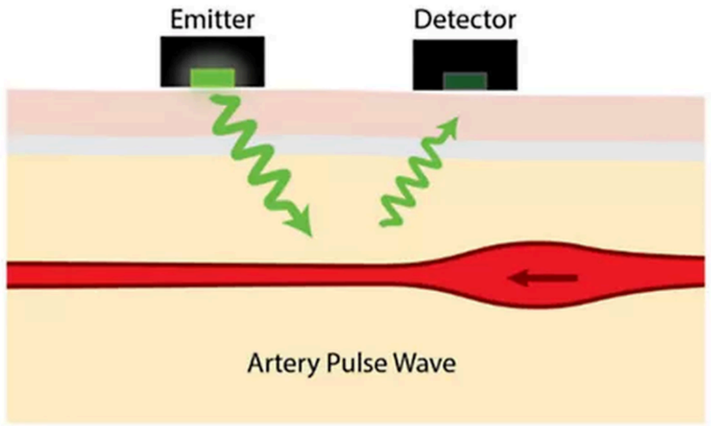
SleepInfoMinutes
1x1 struct
1x1 struct
1x1 struct
1x1 struct
1x1 struct
1x1 struct
1x1 struct
1x1 struct
1x1 struct

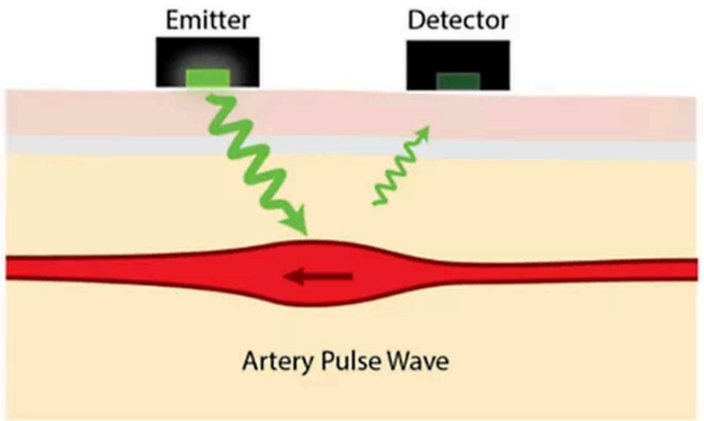
nights(1).SleepInfoMinutes

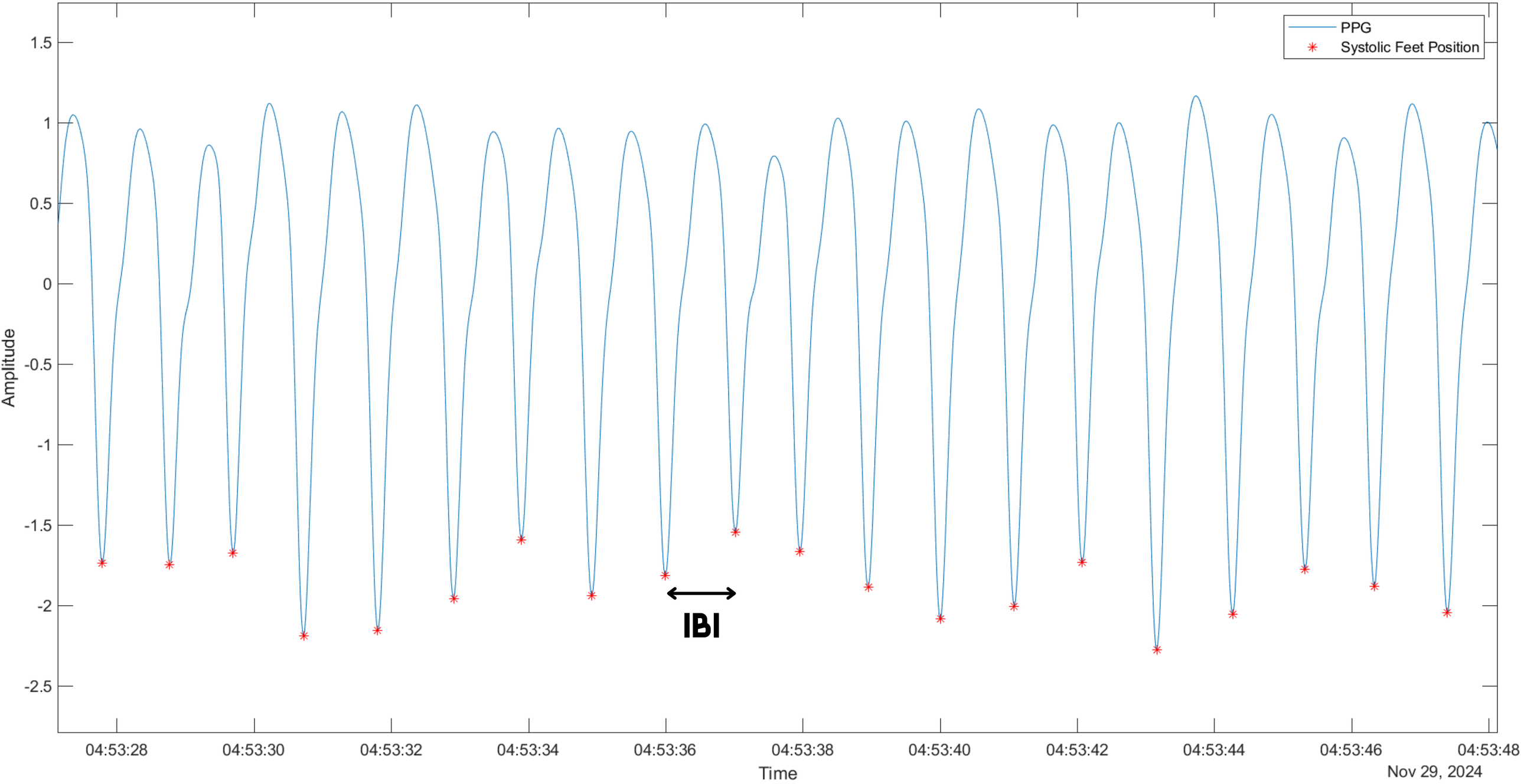
Field	Value
TotalBedTime	525
SleepTime	290
AwakeTime	235

PPG PROCESSING

THE EMPATICA DEVICE IS EQUIPPED WITH A PPG SENSOR THAT WORKS IN REFLECTION MODE USING A GREEN LIGHT SOURCE, THROUGH THIS WE'VE BEEN ABLE TO ACQUIRE IMPORTANT INFORMATION ON THE HRV SUCH AS INTRA BEAT INTERVALS (IBI), RMSD, SDNN AND MEAN HEART RATE






PPG
Systolic Feet Position
Amplitude
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
IBI
04:53:28
04:53:30
04:53:32
04:53:34
04:53:36
04:53:38
04:53:40
04:53:42
04:53:44
04:53:46
04:53:48
Time
Nov 29, 2024

EDA PROCESSING

THE EDA PARAMETERS ARE CHOOSEN ACCORDING TO W. ROMINE AT AL. PAPER, TO DO THAT A PRELIMINARY FILTERING STAGE IS REQUIRED TO OBTAIN THE DESIRED DATA. THIS STEP CONSIST IN A LOW PASS FILTERING WHICH IS DESCRIBED IN THE PAPER OF A.SANO ET AL. FROM 2014.

sensors

Article

Toward Sensor-Based Sleep Monitoring with Electrodermal Activity Measures

William Romine [1], Tanvi Banerjee [2,*] and Garrett Goodman [2]

Int J Psychophysiol. 2014 Dec;94(3):382-9. doi: 10.1016/j.ijpsycho.2014.09.011. Epub 2014 Oct 5.

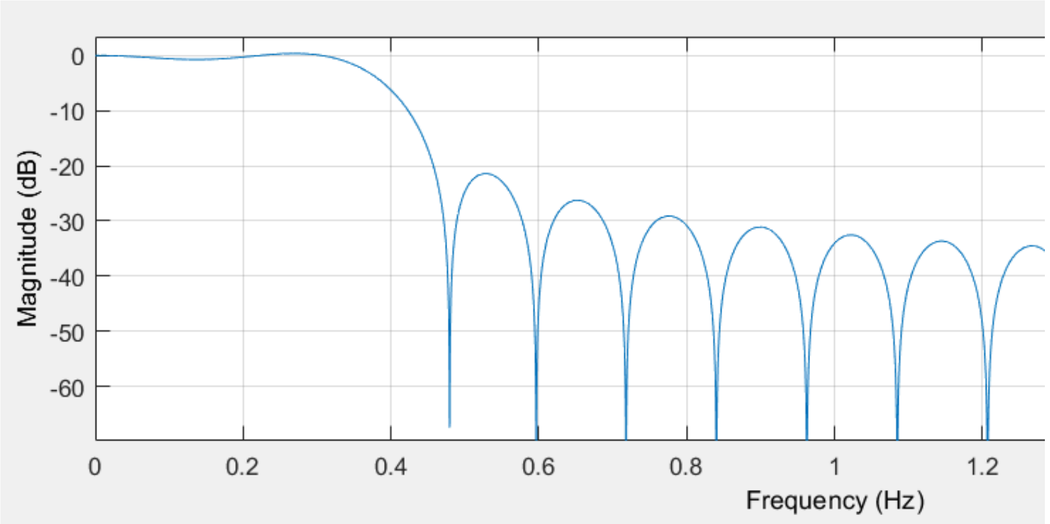
Quantitative analysis of wrist electrodermal activity during sleep

Akane Sano [1], Rosalind W Picard [2], Robert Stickgold [3]

EDA PROCESSING

THE EXTRACTED PARAMETERS ARE:

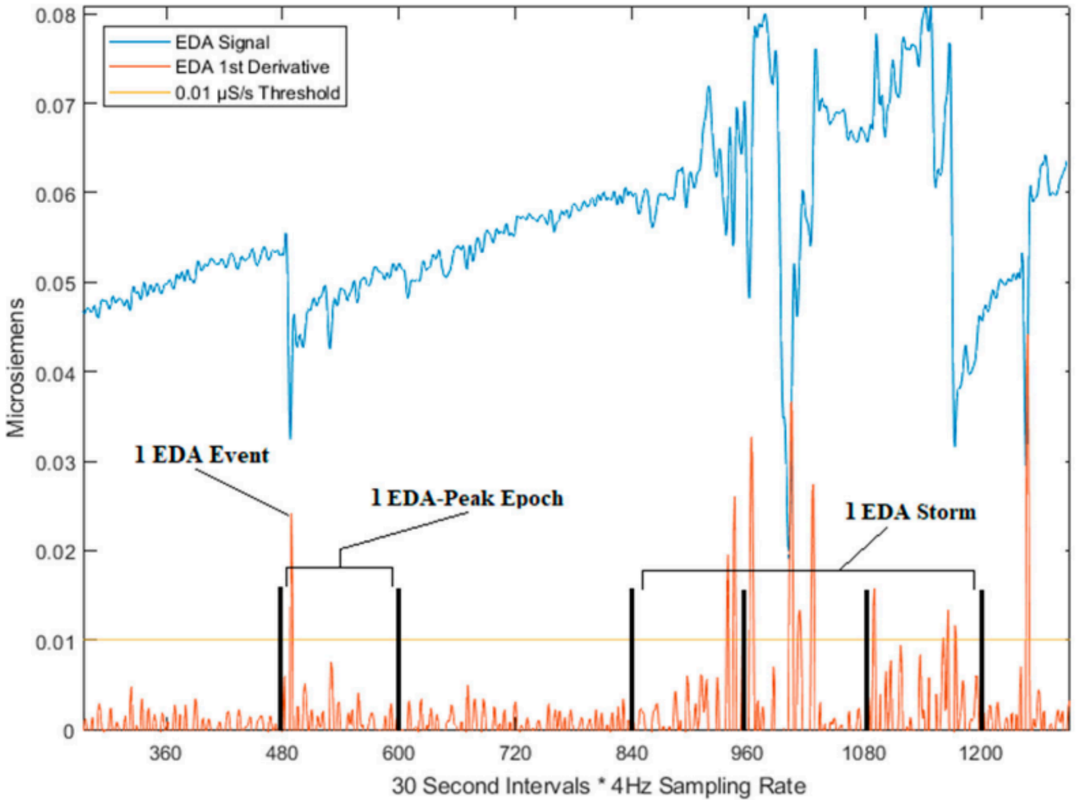
- EDA EPOCHS
- EDA STORMS
- AVERAGE SIZE AND STD OF EDA STORMS
- LARGEST EDA STORM
- NUMBER OF EDA EVENTS

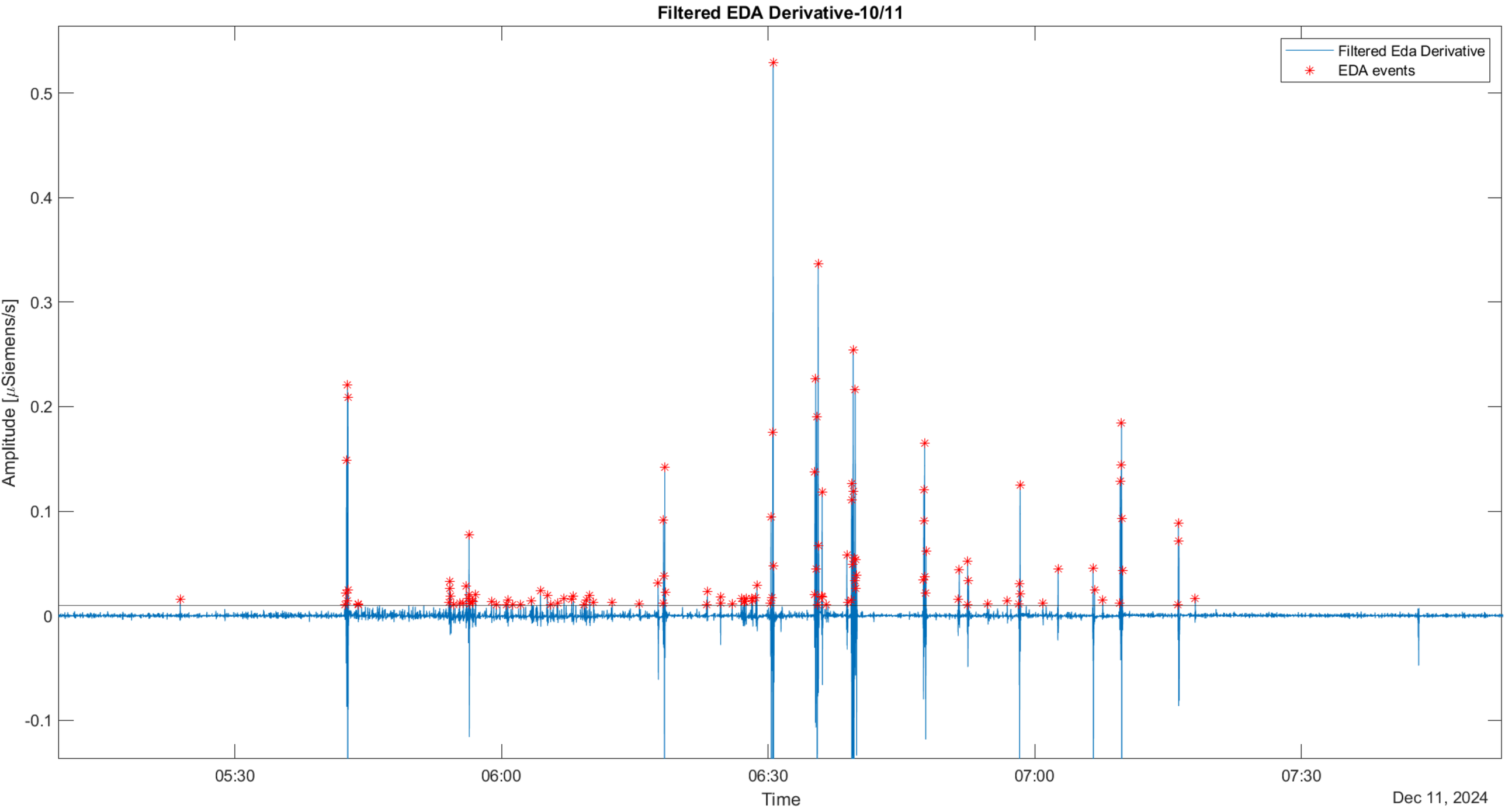


WHILE INFOS SUCH SLEEP DURATION AND EFFICIENCY ARE ALREADY OBTAINED THROUGH THE ACCELERATION PROCESSING

EDA STORMS

AN EDA EPOCH IS DEFINED AS ANY 30 SECONDS TIME INTERVAL IN WHICH THERE IS AT LEAST ONE EDA EVENT, THE SEQUENCE OF AT LEAST TWO CONSECUTIVE EDA EPOCHS IS INSTEAD DEFINED AS AN EDA STORM.

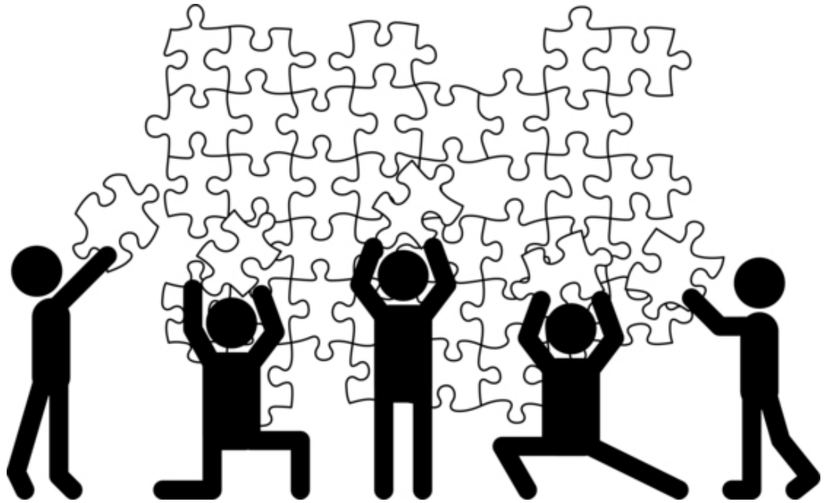



Filtered EDA Derivative-10/11
Filtered Eda Derivative
EDA events
Amplitude [μSiemens/s]
0.5
0.4
0.3
0.2
0.1
0
-0.1
05:30
06:00
06:30
07:00
07:30
Time
Dec 11, 2024

PCA RESULTS

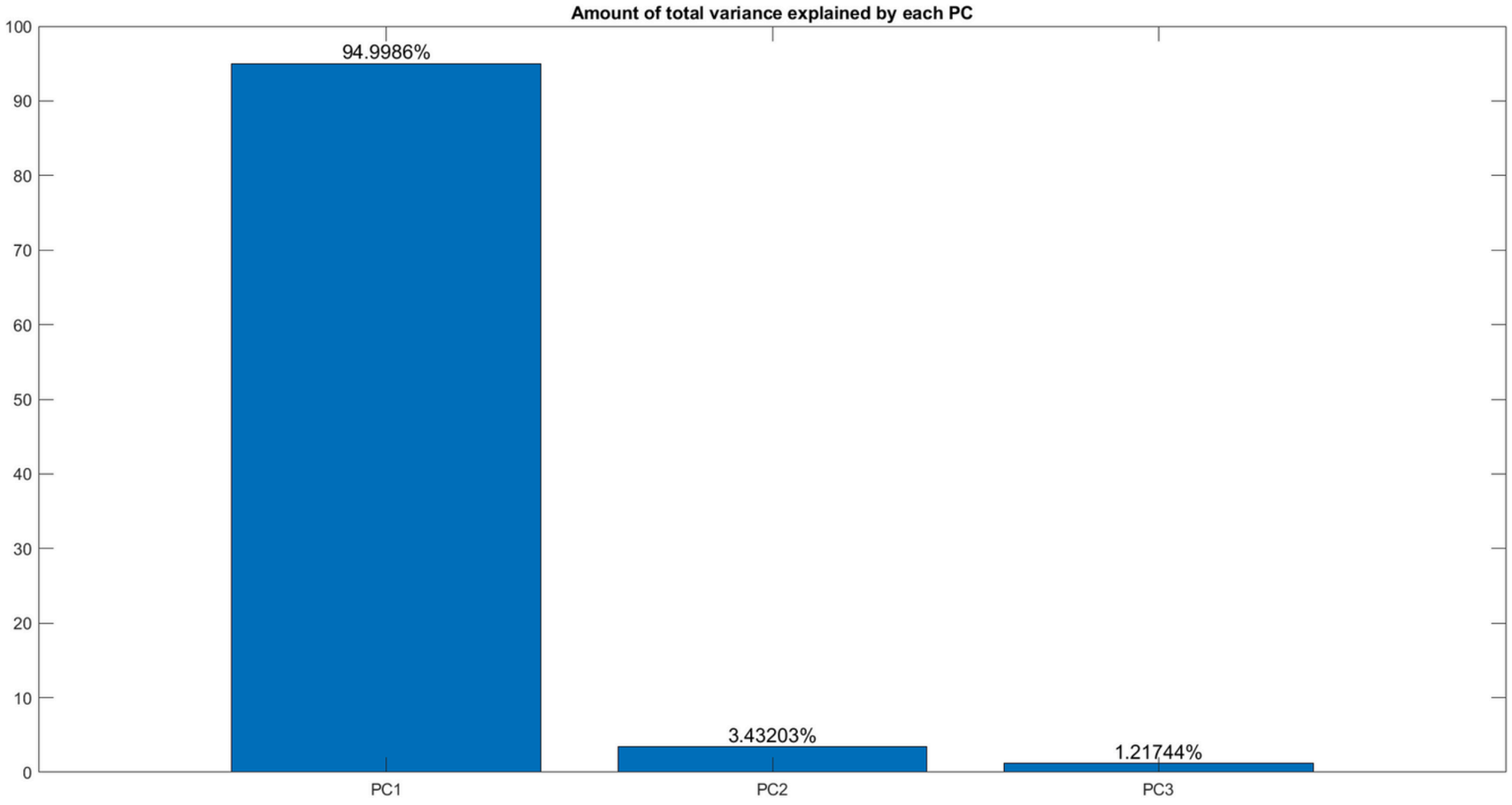
IN ORDER TO ANALYSE ALL THE DIFFERENT DATA A PCA IS BEEN CARRIED OUT TAKING INTO ACCOUNT 9 DIFFERENT VARIABLES:

- AVERAGE EDA STORM SIZE
- STD STORM SIZE
- LARGEST STORM SIZE
- SLEEP EFFICIENCY
- SDNN
- RMSD
- MEAN HEART RATE
- SLEEP QUALITY (PERSONAL OPIONION, FROM 0 TO 10)
- COUNTDOWN TO THE EXAM

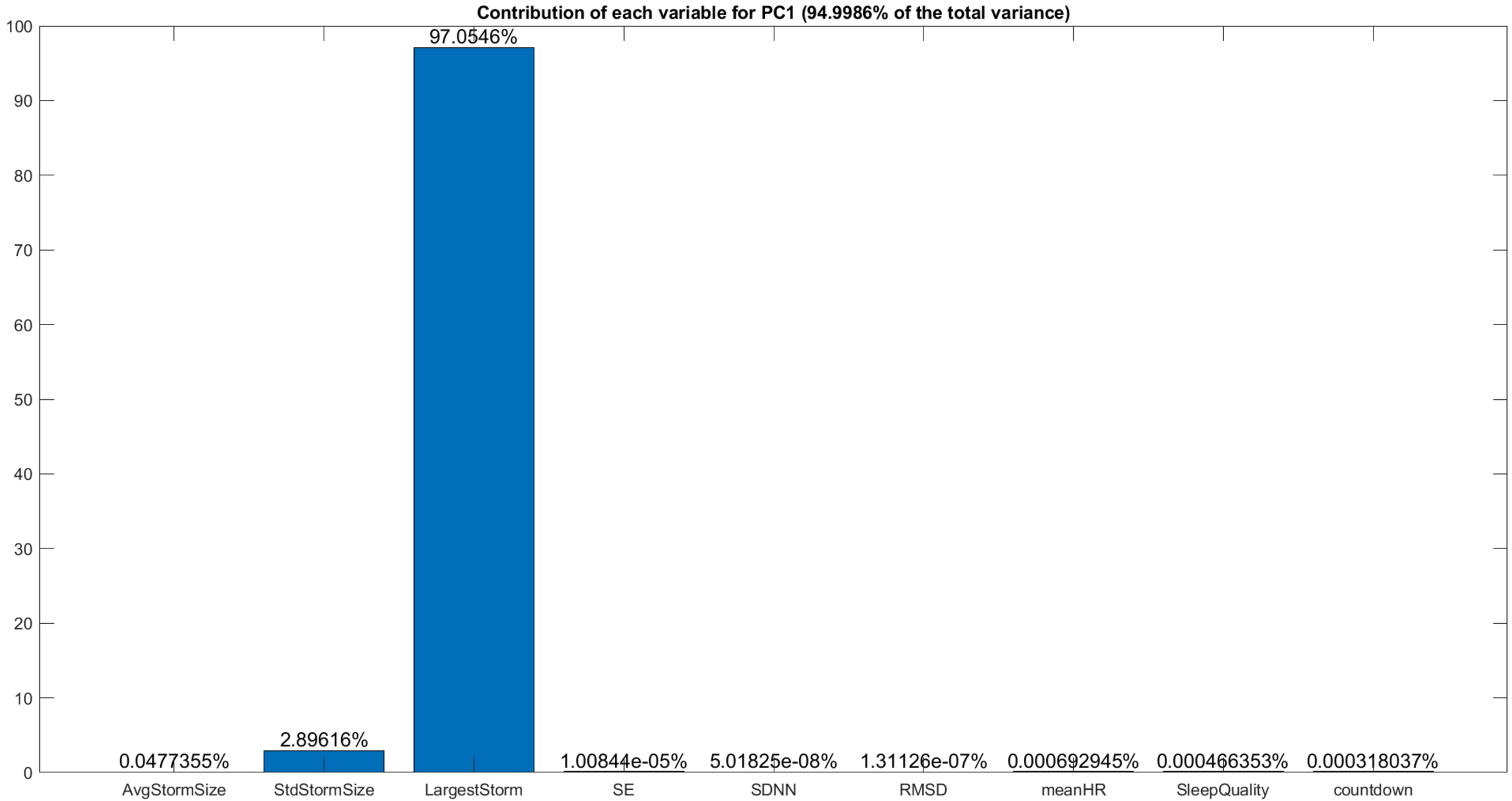


PCA RESULTS

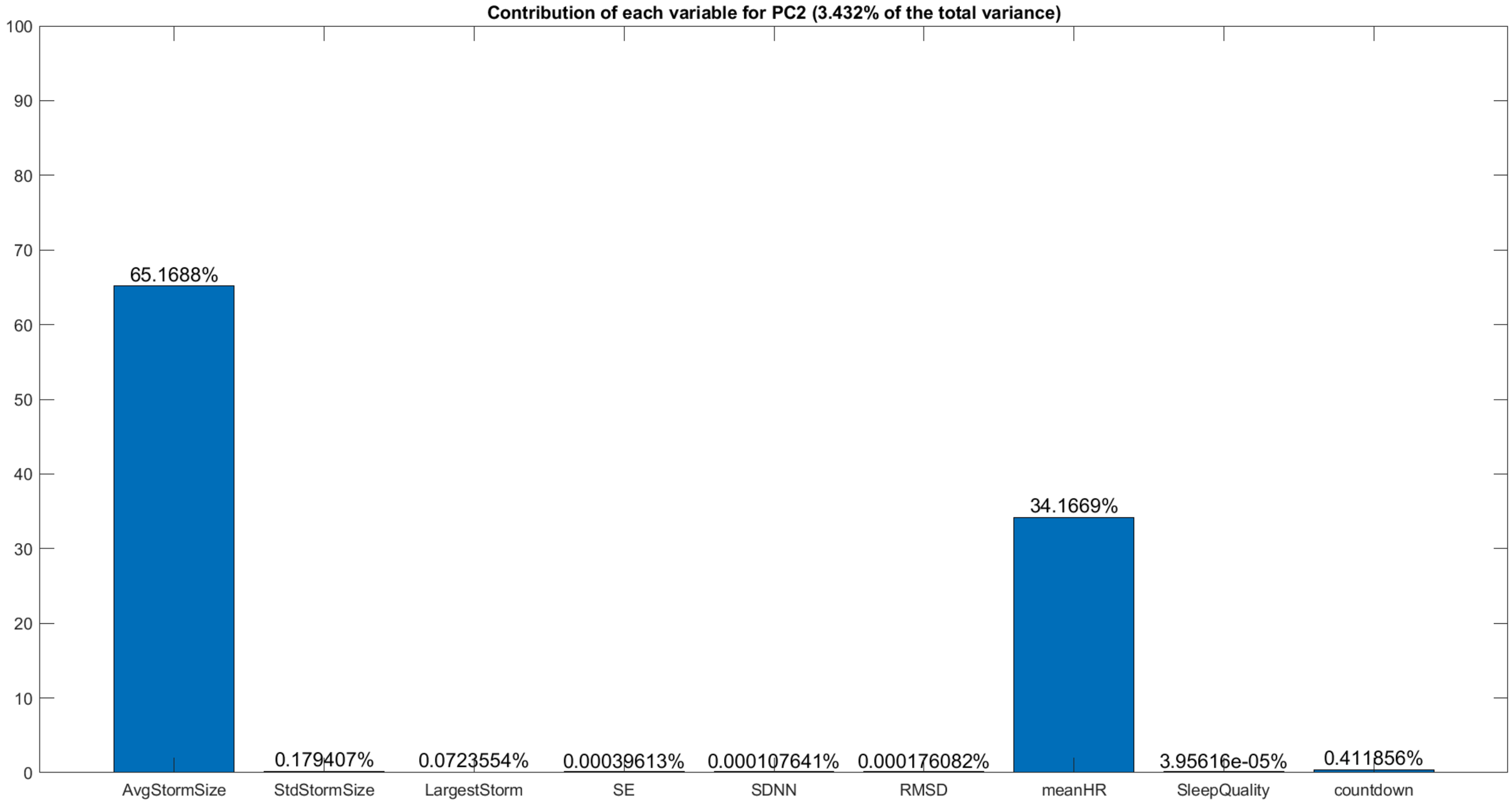
THROUGH THE PCA A SET OF 8 PRINCIPAL COMPONENT HAVE BEEN DISCOVERED BUT ONLY 3 OF THESE EXPRESS AT LEAST 1% OF THE TOTAL DATA VARIANCE.



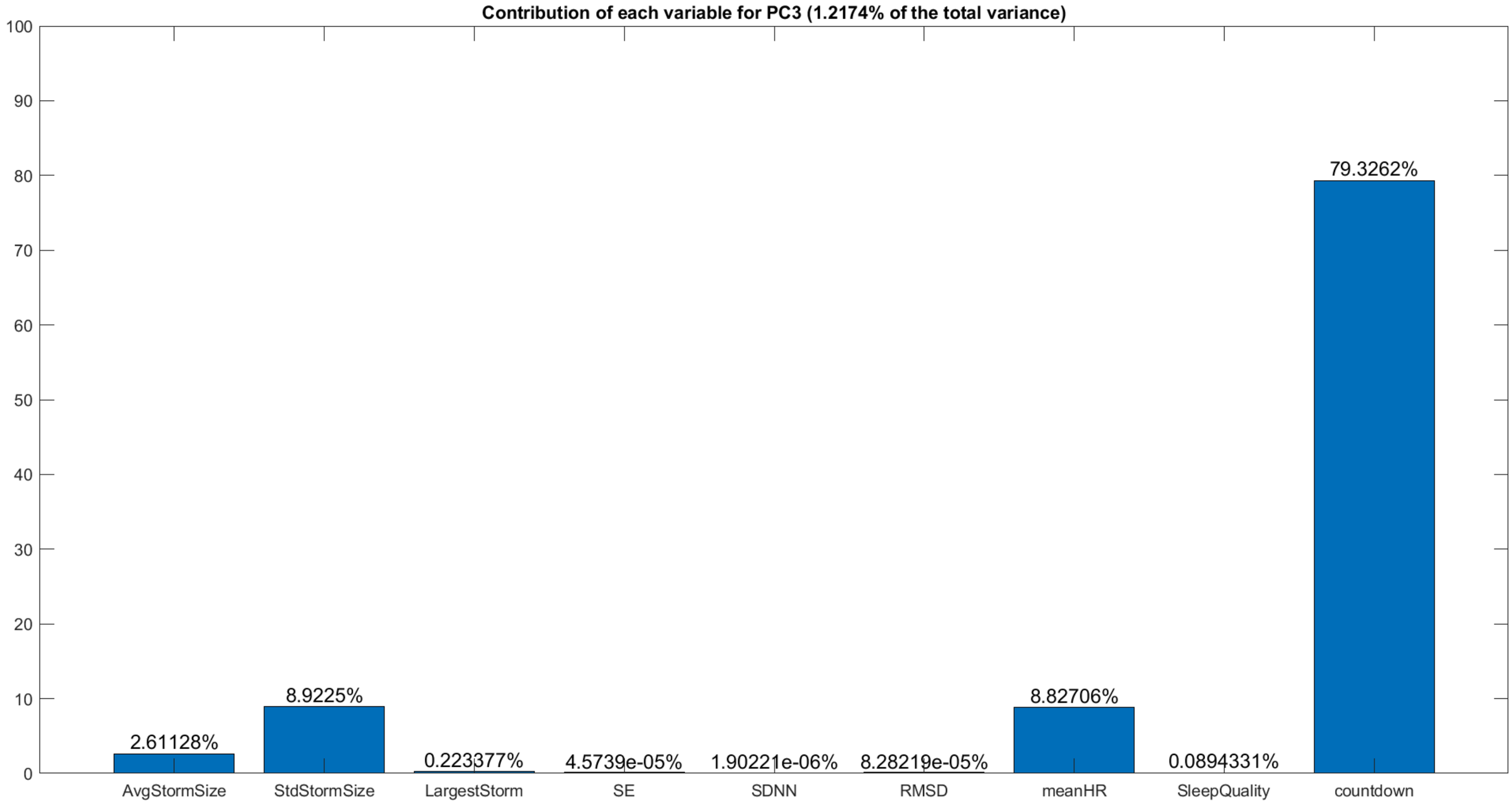
PC1 CONTRIBUTIONS



PC2 CONTRIBUTIONS



PC3 CONTRIBUTIONS



CONCLUSION

WE ANALYZED THE DATA THROUGH AN UNSUPERVISIONED STATISTICAL PROCEDURE THAT REQUIRES STRONG ASSUMPTION SUCH AS THE LINEARITY OF THE DATA WE ARE WORKING ON.

DURING THIS ANALYSIS SUCH PROCEDURE IS BEEN APPLIED EVEN NOT RESPECTING SUCH STRONG CONDITIONS, OBTAINING HIGH LEVEL OF VARIANCE EXPRESSION IN THE FIRST PCA DUE TO AN HIGH CONTRIBUTION OF THE "LARGEST STORM ", WHICH IS ONE OF THE FEATURES THAT MAINLY DESCRIBE ONE OF THE TWO IMPORTANT LATENT FACTOR FOUND IN THE PAPER.

END